

The phase taxonomy of pitch-class set invariants

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Abstract

Working over the chromatic universe $\mathbb{Z}_{12}$, represent a pitch-class *set* A by its 0/1 indicator and take its discrete Fourier transform $\hat{A} = \mathcal{F}(\text{ind } A)$. Write $\hat{A} = |\hat{A}| e^{i\varphi}$. We record the following exact organization: every classical pc-set invariant *we examine* sorts cleanly by **how much of the phase φ it retains**. The interval vector, Babbitt’s hexachord theorem, homometry / the Z-relation, deep scales, the rhythmic-oddity property, and common tones under transposition are all functions of the phase-**blind** power spectrum $|\hat{A}|^2$ alone — equivalently the autocorrelation, equivalently (in crystallography) the **Patterson function**. The sum / index vector — common tones under inversion — is a function of the squared coefficient $\hat{A}^2$, which retains only *partial* phase and, as an invariant of the set class, is weak. Then comes the twist: the third-order **triple correlation** (whose transform is the bispectrum $\hat{A} \cdot \hat{A} \cdot \overline{\hat{A}}$) is *also* phase-blind, yet over $\mathbb{Z}_{12}$ it **resolves the Z-relation completely** — separating homometric chords needs not more phase but higher *order*. The full $\hat{A}$ (magnitude **and** phase) is, last, a complete invariant determining the set exactly. The boundaries of this ladder of increasing resolving power are machine-checked in `Fourier.lean`: the phase-blind collapse (`homometric_iff_normSq_dft_eq`, `allIntervalTetrachords_homometric`), the third-order resolution (`triple_tpose_invariant`, `carterPair_triple_distinct`), and the complete-invariant capstone (`Ahat_injective`), in one file with a clean axiom audit. The crystallographic crossing is rigorous, not analogy: autocorrelation = Patterson function, $|\hat{A}|^2$ = X-ray diffraction intensity, the Z-relation = the phase problem. Every ingredient is classical (the triple correlation’s resolving power is the bispectrum’s recovery of the Fourier phase up to a ramp); the contribution is the unified taxonomy as one organization with its boundaries machine-checked — a first formalization, not new mathematics.

1 Two chords that measure the same

In 1973 Allen Forte, cataloguing the chords of atonal music, marked certain pairs with the letter **Z** [1]: distinct set classes — not related by transposition or inversion — that nonetheless share the very same interval vector, the basic tally of how many intervals of each size a chord contains. Forte’s *Z-relation* named a puzzle David Lewin had already met in 1960 [2] and Howard Hanson had called

*Prepared with the assistance of an AI system (Claude, Anthropic). The mathematics is classical; every claim, scope statement, and limitation is the author’s responsibility. The boundary of the contribution is stated plainly in Section 9, and the Lean 4 kernel—not the assistant—is what certifies the formalized parts.

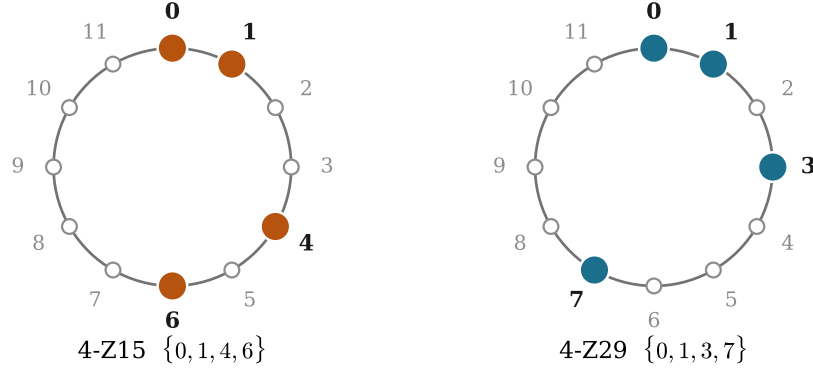


Figure 1: The canonical homometric (Z-related) pair on the pitch-class clock (0 at twelve o’clock, clockwise; filled nodes are members): 4-Z15 $\{0, 1, 4, 6\}$ and 4-Z29 $\{0, 1, 3, 7\}$. Both share the interval vector — and the power spectrum $|\hat{A}|^2 = [16, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2]$ — yet are *not* T/I -related: the same magnitude, a different phase (§3).

isomeric [3]: two chords can be indistinguishable to interval analysis and yet be genuinely different objects. The canonical example is a pair of all-interval tetrachords, $\{0, 1, 4, 6\}$ and $\{0, 1, 3, 7\}$: each contains all six interval classes exactly once, so their interval vectors are identical, but no transposition or inversion carries one to the other.

What is that shared descriptor, mathematically? **The interval vector is the magnitude of a Fourier transform with its phase thrown away.** Encode a chord as a 0/1 indicator on the twelve pitch classes $\mathbb{Z}_{12}$ and take its discrete Fourier transform $\hat{A}$; the interval vector is a function of $|\hat{A}|^2$, the squared magnitudes alone. **Two chords are Z-related precisely when they share $|\hat{A}|^2$ but disagree in the phase of $\hat{A}$** — the part $|\hat{A}|^2$ cannot see. The phase is the hidden variable of this note. (The same magnitude-without-phase loss is, verbatim, the phase problem of X-ray crystallography: the interval vector is the *Patterson function* of 1934 and homometric crystal structures [5] are Z-related chords — a rigorous crossing we make precise in §3, after Mandereau et al. 2011 [6]. We note it and move on; the subject here is the music and its mathematics.)

Note #1 of this series followed one *chord* — Stravinsky’s *Petrushka* collection, the set class 6-30 — from a tritone into group theory. This note follows one *question* — why two chords can measure the same — into the Fourier transform, and finds that the phase sorts the whole catalogue of pitch-class set invariants into a single **ladder**: the interval vector and its relatives at the phase-blind bottom, the complete set class at the top, each invariant placed by how much of the set it pins down — a climb with a twist, since the phase is not the only currency.

2 Setup: the DFT of a pitch-class set

Pitch classes are $\mathbb{Z}_{12}$. A pc-set $A \subseteq \mathbb{Z}_{12}$ is encoded by its indicator $\text{ind } A : \mathbb{Z}_{12} \rightarrow \mathbb{C}$, $\text{ind } A(j) = 1$ if $j \in A$ else 0 (`Fourier.lean` line 29). Its discrete Fourier transform is $\hat{A} = \mathcal{F}(\text{ind } A)$, built on Mathlib’s `ZMod.dft` (David Loeffler, Apache-2.0; line 32). In collapsed form (line 35),

$$\hat{A}(t) = \sum_{a \in A} \text{stdAddChar}(-(a \cdot t)) = \sum_{a \in A} e^{-2\pi i at/12}.$$

Mathlib’s sign convention is $\text{stdAddChar } j = e^{2\pi i j/12}$ and $\mathcal{F}\Phi(k) = \sum_j \text{stdAddChar}(-(jk)) \Phi(j)$; the DC coefficient is the cardinality, $\hat{A}(0) = |A|$ (`Ahat_zero`, line 42).

Each coefficient decomposes as $\hat{A}(t) = |\hat{A}(t)| e^{i\varphi(t)}$. Two derived objects carry the rungs of the taxonomy. The **power spectrum** $\text{powerSpec}(t) = \hat{A}(t) \overline{\hat{A}(t)} = |\hat{A}(t)|^2$ (line 88) discards the phase entirely. The **sum spectrum** $\text{sqSpec}(t) = \hat{A}(t) \cdot \hat{A}(t) = \hat{A}(t)^2$ (line 469) is the coefficient *squared* — not its squared magnitude — and *retains* the phase (`sqSpec_eq_sq`, line 473).

The set *class* carries two phase symmetries. Transposition is a frequency-linear phase **ramp**: $\hat{A}(T_k A)(t) = \text{stdAddChar}(-(kt)) \hat{A}(t)$ (`Ahat_tpose`, line 51), i.e. $\hat{A}(t) \mapsto e^{-2\pi i k t / 12} \hat{A}(t)$. Inversion is **conjugation plus a ramp**. Passing from a set to its T/I class quotients out exactly these phase operations; $|\hat{A}|$ is invariant under the ramp, but the phase still carries information $|\hat{A}|$ cannot — this is the whole story below.

3 The phase-blind floor: $|\hat{A}|^2 = \text{autocorrelation} = \text{Patterson}$

P₀ — the hook, made precise. The two tetrachords of §1, $\{0, 1, 4, 6\}$ (Forte 4-Z15) and $\{0, 1, 3, 7\}$ (Forte 4-Z29), are identical to interval analysis yet not T/I -related — the Z-relation. We now locate exactly *what* invariant fails to separate them, and why: **it is a function of $|\hat{A}|^2$ alone, and the answer is the phase.**

To see it, read the interval vector spectrally. The ordered autocorrelation $\text{autocorr}(A, d) = \#\{(a, b) \in A \times A : a - b = d\}$ (line 84) is the inverse DFT of the power spectrum (Wiener-Khinchin), $\text{autocorr}(A, d) = \mathcal{F}^{-1}(|\hat{A}|^2)(d)$ (`autocorr_eq_invDFT`, line 104); the raw interval vector is this autocorrelation regraded by interval class, $\text{IVraw}(A, k) = \sum_{\text{ivc } d=k} \mathcal{F}^{-1}(|\hat{A}|^2)(d)$ (`IVraw_eq_invDFT_power`, line 167). The interval vector is therefore a function of $|\hat{A}|^2$ alone: it cannot see the phase.

Homometry = the kernel of “forget the phase.” Two pc-sets are **homometric** when they share the same autocorrelation (`Homometric`, line 353). Exactly: homometric iff equal power spectrum (`homometric_iff_powerSpec_eq`, line 358), equivalently iff equal $|\hat{A}|^2$ at every frequency (`homometric_iff_normSq_dft_eq`, line 373). **Homometric $\iff$ same $|\hat{A}|^2$: the equivalence “same magnitude, any phase,” the information it discards being the phase of $\hat{A}$.**

The crystallographic crossing (rigorous, not analogy). In X-ray crystallography the autocorrelation of an electron density is the **Patterson function** (Patterson 1934 [4]); the measured diffraction intensity is $|\hat{A}|^2$, the squared structure-factor magnitude; and the central obstruction — recovering a structure from its diffraction pattern — is the **phase problem**, because distinct densities can share $|\hat{A}|^2$. **The Z-relation is that phase problem on $\mathbb{Z}_{12}$: same diffraction, different phase** — autocorrelation = Patterson, Z-relation = phase problem. The correspondence Z-relation $\leftrightarrow$ Patterson is exactly Mandereau et al. 2011 [6]. In `Fourier.lean` the Patterson function is named as such (`Patterson`, line 349) and the phase-problem theorem is `homometric_iff_normSq_dft_eq`.

P₀ answered — the witness. The all-interval tetrachords are the canonical homometric pair (Figure 2): Homometric $\{0, 1, 4, 6\} \{0, 1, 3, 7\}$ (`allIntervalTetrachords_homometric`, line 390, by `decide`). Verified in Sage (container `cognitive_arch_sage`): both sets have autocorrelation $[4, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1]$ and power spectrum $|\hat{A}|^2 = [16, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4]$, equal at every frequency; their prime forms $(0, 1, 4, 6) \neq (0, 1, 3, 7)$ are not T/I -related (Figure 1) [b audio]. Same $|\hat{A}|^2$, different set: the floor of the taxonomy.

P₁. Is the interval vector alone in this blindness, or does the whole floor share it?

Every phase-blind invariant is a function of $|\hat{A}|^2$: the interval vector (`IVraw_eq_invDFT_power`); Babbitt’s hexachord theorem — a hexachord and its complement share $|\hat{A}|$ off $t = 0$

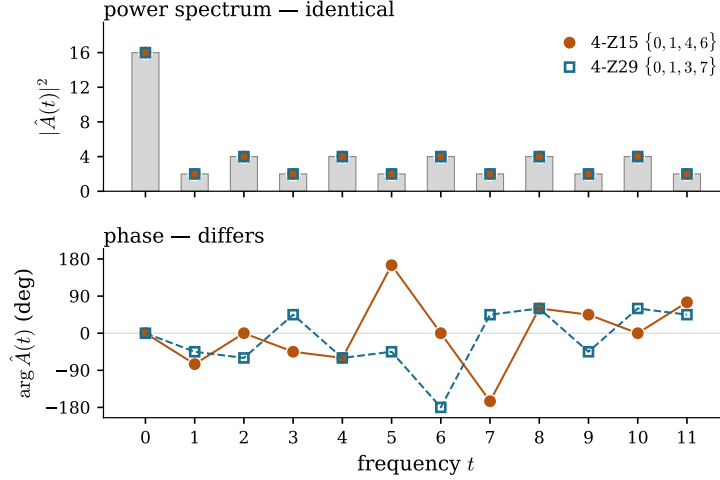


Figure 2: The two all-interval tetrachords share the power spectrum $|\hat{A}(t)|^2$ exactly (top — the bars coincide), yet their phases $\arg \hat{A}(t)$ differ at most frequencies (bottom). Every phase-blind invariant is a function of $|\hat{A}|^2$ alone, so none can separate them; the phase is the difference the interval vector cannot see.

(`hexachord_abs_eq`, line 327), hence the same interval vector; common tones under transposition $\#(A \cap T_k A) = \mathcal{F}^{-1}(|\hat{A}|^2)(k)$ (`commonTones_eq_invDFT`, line 223); deep scales — the interval-vector entries are the transposition common-tone counts, all distinct (`IsDeep_iff_commonTones_nodup`, line 304); the rhythmic-oddity property — the autocorrelation vanishes at the tritone (`rop_iff_autocorr_zero`, line 580). All of these collapse on the Z-pair, because all of them are read off $|\hat{A}|^2$.

Over the 222 nontrivial set classes of $\mathbb{Z}_{12}$ there are **23 Z-pairs** (every Z-group has size 2), covering 46 set classes: 1 tetrachord pair (the all-interval tetrachords 4-Z15 / 4-Z29), 3 pentachord, 15 hexachord (the classic hexachordal Z-relations), 3 septachord, 1 octachord (`sage/phase_taxonomy.sage`, `sage/sc_count.sage`).

4 The partial-phase rung: $\hat{A}^2 = \text{the sum} / \text{index vector}$

P₂. Can any invariant see the phase the interval vector misses?

Where the interval vector counts tones held under *transposition* (the *difference* pairing $a - b$, governed by $|\hat{A}|^2$), the **sum / index vector** counts tones held under *inversion* $I_j(x) = j - x$ (the *sum* pairing $a + b$). The held count is $\text{commonTonesInv}(A, j) = \#(A \cap I_j A)$ (line 462), equal to the sum vector $\text{sumcorr}(A, j) = \#\{(a, b) : a + b = j\}$ (`commonTonesInv_eq_sumcorr`, line 508).

Its transform is $\hat{A}^2$, not $|\hat{A}|^2$. The sum vector is the inverse DFT of the *squared coefficient*, $\text{sumcorr}(A, j) = \mathcal{F}^{-1}(\hat{A}^2)(j)$ (`sumcorr_eq_invDFT`, line 515), with $\text{sqSpec}(t) = \hat{A}(t)^2$ (`sqSpec_eq_sq`, line 473). Because squaring keeps the coefficient rather than multiplying by its conjugate, $\hat{A}^2$ retains phase that $|\hat{A}|^2 = \hat{A} \cdot \overline{\hat{A}}$ destroys. The sum vector is therefore **phase-dependent**.

It splits the homometric pair $|\hat{A}|^2$ cannot. About axis $j = 0$, the two all-interval tetrachords hold different numbers of tones: `commonTonesInv {0,1,4,6} 0` $\neq$ `commonTonesInv {0,1,3,7} 0` (`sumVector_distinguishes_homometric`, line 555, by `decide`). Verified (`sage/isum.sage`): at $j = 0$, `{0,1,4,6}` [b audio] holds **2** tones, `{0,1,3,7}` [b audio] holds **1**. **The phase the interval vector cannot see, the sum vector sees:** $\hat{A}^2$ retains partial phase and splits the Z-pair.

Honest: partial, not all. $\hat{A}^2$ retains *some* phase, not all of it. The full sum vectors of the pair are `[2,2,1,0,2,2,2,2,1,0,2,0]` and `[1,2,2,2,2,0,1,2,2,0,2,0]` (`sage/isum.sage`). These differ *as indexed vectors* — axis j against axis j — but they are **permutations of each other**: both sort to `[0,0,0,1,1,2,2,2,2,2,2,2]`. So the *indexed* sum vector (per axis) sees the phase; the bare *multiset* of held-tone counts does not. We state the distinction as indexed, never as a global multiset difference.

5 The third-order rung: the triple correlation resolves homometry

Must we recover the lost phase at all to tell the homometric pair apart?

Surprisingly, no. The sum vector of the previous section saw phase only at a fixed axis and is weak as a class invariant; the object that resolves homometry completely keeps *no* phase at all — it climbs one *order* instead. That higher-order object is the **triple correlation** `triple A j k = # (A ∩ TjA ∩ TkA)` — the number of pitch classes x with $x, x - j, x - k$ all in A — whose DFT is the **bispectrum** $\hat{A}(t_1) \hat{A}(t_2) \hat{A}(t_1 + t_2)$, the third-order analog of `autocorr =  $\mathcal{F}^{-1}(|\hat{A}|^2)$` . It is **phase-blind in the diffraction sense yet transposition-invariant** (`triple_tpose_invariant`), and, unlike the interval vector, it **distinguishes the homometric pair**: `triple {0,1,4,6}  $\neq$  triple {0,1,3,7}` (`carterPair_triple_distinct`, by `decide`). Over $\mathbb{Z}_{12}$ it separates **all 23 homometry (Z-related) pairs and all 222 set classes** (exhaustive computation, two independent ways, 0 collisions). It is **not** inversion-invariant (`triple_not_inversionInvariant`) — it tells a set from its inversion — which is exactly why, folded by inversion, it is the complete set-class invariant of the next section. The mathematics is classical: the bispectrum recovers the Fourier phase up to a linear ramp (the triple-correlation completeness theorem, Kakarala [11]; the bispectrum of Tukey and Brillinger [10]; the crystallographic higher-order Patterson function), the $\mathbb{Z}_n$ reconstruction-from-3-deck question is Keleti–Kolountzakis 2006 [12], and the transposition-invariant coefficient products are Yust–Amiot 2022 [9]. Our contribution is the exact $\mathbb{Z}_{12}$ statement and its first formalization (Brick 22).

6 The complete invariant: the full $\hat{A}$

P₃. What would it take to see all of the phase?

The full DFT $\hat{A}$ (magnitude and phase) determines the pitch-class set exactly: $\hat{A}_A = \hat{A}_B \Rightarrow A = B$ (`Ahat_injective`, line 427), with biconditional `Ahat_eq_iff` (line 431). The proof is short and classical: `ZMod.dft` is a `LinearEquiv`, hence injective (`dft_injective`), and the indicator is injective because membership is recoverable as `ind A (x) = 1` using `(1 :  $\mathbb{C}$ )  $\neq$  0` (`ind_injective`, line 418). The phase is exactly the data that completes the interval vector to a total invariant.

From SET to set CLASS — the computational boundary. `Ahat_injective` proves $\hat{A}$ determines the set; the set-class completeness (modulo the T/I phase symmetries) is the computational statement, not a Lean theorem. Quotienting by the two phase symmetries

of §2 — the transposition ramp and the inversion conjugation-plus-ramp — gives a candidate complete invariant of the set class. Verified (`sage/phase_taxonomy.sage`): take the canonical signature = lex-min of $\hat{A}$ over the 24-element phase-symmetry group ($12 \text{ ramps} \times \{\text{id}, \text{conj}\}$); the sweep finds **0 collisions**, distinct signatures = $224 = \text{number of } T/I \text{ orbits} (= 222 \text{ nontrivial set classes} + \text{the empty set} + \text{the chromatic aggregate})$. So the full $\hat{A}$ modulo $\{\text{ramp}, \text{conj-ramp}\}$ is a complete set-class invariant — and homometry, the phase-blind collapse, sits exactly one rung below this completeness boundary.

7 The taxonomy, and the loop back to 6-30

The whole map, with each invariant pinned to its slice of $\hat{A}$:

Invariant	Slice of $\hat{A}$	Phase	Complete?	Lean (<code>Fourier.lean</code>)
Interval vector (B.1)	$ \hat{A} ^2$	blind	no [†]	<code>IVraw_eq_invDFT_power</code> (L167)
Hexachord theorem (B.2)	$ \hat{A} $	blind	—	<code>hexachord_abs_eq</code> (L327)
Homometry / Z-relation (B.3)	$ \hat{A} ^2$	blind	collapses	<code>homometric_iff_normSq_dft_eq</code> (L373)
Common tones / T_k (O.1)	$ \hat{A} ^2$	blind	no	<code>commonTones_eq_invDFT</code> (L223)
Deep scale (L.4)	$ \hat{A} ^2$	blind	no	<code>IsDeep_iff_commonTones_nodup</code> (L304)
Rhythmic oddity (N.1)	$ \hat{A} ^2$ at $t = 6$	blind	no	<code>rop_iff_autocorr_zero</code> (L580)
Sum / index vector (O.2)	$\hat{A}^2$	partial	no (weak)	<code>sumcorr_eq_invDFT</code> (L515)
Triple correlation (B22)	$\hat{A} \cdot \hat{A} \cdot \overline{\hat{A}}$	blind	resolves	<code>triple_tpose_invariant</code> (L703)
Full DFT	$\hat{A}$	complete	YES[‡]	<code>Ahat_injective</code> (L427)

Table 1: Each classical invariant pinned to the slice of $\hat{A}$ it retains. [†]Homometry collapses the interval vector (§3). [‡]The full $\hat{A}$ determines the *set* in Lean; the *set-class* completeness (mod the T/I phase symmetries) is the computational statement of §6.

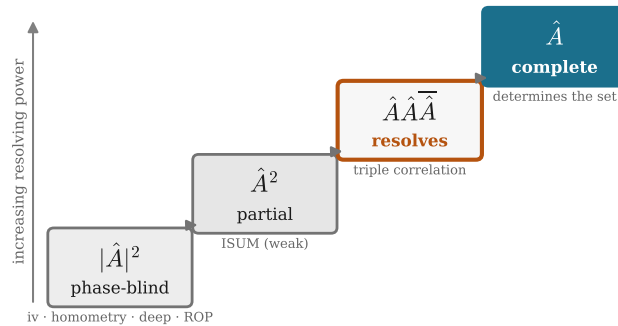


Figure 3: The taxonomy as a ladder of increasing information. $|\hat{A}|^2$ is phase-blind and homometry collapses it; $\hat{A}^2$ retains only partial (weak) phase; the triple correlation (§5) is the phase-blind third-order invariant that *resolves* homometry; the full $\hat{A}$ sees everything (`Ahat_injective`). The top rung is the completeness boundary, highlighted as in Table 1.

P₄. What does this say back to the tritone self-duality of Note #1?

Note #1 [17] identified $6\text{-}30 = (013679)$ as the unique nontrivial $\mathbb{Z}_{12}$ set class carrying a tritone-centered self-duality: its T/I -stabilizer is exactly the center $\{T_0, T_6\}$. That self-duality is a **phase fact**. The condition $T_6 \in \text{Stab}$ is equivalent to $\hat{A}$ being supported on the even frequencies (vanishing on the odd ones) — now machine-witnessed: **0 mismatches over all 4096 subsets** of $\mathbb{Z}_{12}$ (`sage/phase_taxonomy.sage`), the strongest universe, not merely the set classes. But self-duality needs *more* than tritone symmetry: it also requires an inversion-free phase condition that $|\hat{A}|$ cannot see. This is the H2 thread of Note #1, where self-duality was **refuted as a clean $|\hat{A}|^2$ theorem** — Fourier explains the phase *requirement*, it does not deliver self-duality from the magnitude alone. The taxonomy retro-explains why Note #1’s result lived **above** the interval-vector floor: a fact about tritone symmetry and self-duality is a fact about the *phase* of $\hat{A}$, invisible to the phase-blind column where the interval vector and all its relatives sit.

8 What is machine-checked

P₅. Can we trust all this?

The ladder is checked by the Lean 4 kernel (Table 2), namespace `Fourier`, sorry-free. The axiom audit at the foot of `Fourier.lean` (`#print axioms`, lines 727–766) returns at most `[propext, Classical.choice, Quot.sound]` with no `sorryAx` on every theorem; the decidable witnesses (`sumVector_distinguishes_homometric`, `carterPair_triple_distinct`, `triple_not_inversionInvariant`, `diatonic_isDeep`, `augmented_not_isDeep`) are even choice-free (`[propext, Quot.sound]`), confirmed by recompiling the file in the `godsil` fast-loop environment (EXIT 0).

`Ahat_injective` proves $\hat{A}$ determines the **set**. The set-**class** completeness (modulo the T/I phase symmetries) is **computational** (the Sage sweep, 0 collisions over 224 T/I orbits), not a Lean theorem — the same posture as Note #1, where Lean proves the engine and Sage gives the specific sweep. The witnesses (Table 3) are re-runnable.

Script	What it witnesses
<code>phase_taxonomy.sage</code>	Homometric pair $ \hat{A} ^2$ equal; T_6 -symmetry $\iff$ odd-frequency vanishing, 0 mismatches over all 4096 subsets ; full $\hat{A} \bmod \{\text{ramp}, \text{conj-ramp}\}$ complete, 0 collisions, 224 distinct signatures .
<code>homometry.sage</code>	The all-interval tetrachords share autocorrelation $[4, 1, 1, 1, 1, 2, 1, 1, 1, 1, 1]$ and $ \hat{A} ^2 = [16, 2, 4, 2, 4, 2, 4, 2, 4, 2, 4, 2]$; prime forms differ, not T/I -related.
<code>isum.sage</code>	Sum-vector phase distinction: at $j = 0$, $\{0, 1, 4, 6\}$ holds 2, $\{0, 1, 3, 7\}$ holds 1; the full sum vectors differ as indexed vectors but are permutations of each other.
<code>sc_count.sage</code>	Z-relation census: 23 Z-pairs over 222 nontrivial set classes (1 tetra / 3 penta / 15 hexa / 3 septa / 1 octa).

Table 3: Sage witnesses (container `cognitive_arch_sage`, Sage 10.8), each re-runnable.

Theorem (line)	Statement
<code>IVraw_eq_invDFT_power</code> (167)	Interval vector = $\mathcal{F}^{-1}(\hat{A} ^2)$ summed over each interval class — the phase-blind keystone.
<code>homometric_iff_powerSpec_eq</code> (358)	Homometric $\iff$ equal power spectrum — homometry is the kernel of “forget the phase.”
<code>homometric_iff_normSq_dft_eq</code> (373)	Crystallographic form: homometric $\iff$ equal diffraction intensity $ \hat{A} ^2$ — the phase problem.
<code>allIntervalTetrachords_homometric</code> (390)	Homometric $\{0, 1, 4, 6\} \{0, 1, 3, 7\}$ — the canonical Z-pair, same $ \hat{A} ^2$, not T/I -related.
<code>commonTones_eq_invDFT</code> (223)	Common tones under $T_k = \mathcal{F}^{-1}(\hat{A} ^2)(k)$.
<code>sumcorr_eq_invDFT</code> (515)	Sum / index vector = $\mathcal{F}^{-1}(\hat{A}^2)$ — the phase-dependent face.
<code>sqSpec_eq_sq</code> (473)	$\text{sqSpec}(t) = \hat{A}(t)^2$ — the coefficient squared, retaining phase.
<code>commonTonesInv_eq_sumcorr</code> (508)	Held tones under I_j = the sum vector at j (Lewin/Rahn; first formalization).
<code>sumVector_distinguishes_homometric</code> (555)	The sum vector splits the homometric pair $ \hat{A} ^2$ cannot.
<code>ind_injective</code> (418)	A pc-set is determined by its 0/1 indicator.
<code>Ahat_injective</code> (427)	The capstone. The full DFT determines the set: $\hat{A}_A = \hat{A}_B \Rightarrow A = B$.
<code>Ahat_eq_iff</code> (431)	Biconditional form: $\hat{A}_A = \hat{A}_B \iff A = B$.
<code>triple_tpose_invariant</code> (703)	The triple correlation is transposition-invariant (3rd-order, phase-blind).
<code>carterPair_triple_distinct</code> (713)	The triple correlation distinguishes the homometric pair the interval vector cannot.
<code>triple_not_inversionInvariant</code> (721)	The triple correlation is not inversion-invariant (it tells a set from its inversion).

Table 2: The phase-taxonomy ladder in `Fourier.lean`, machine-checked — the phase-blind collapse, the partial-phase rung, and the complete-invariant capstone in one file.

9 Novelty, scope, and what this note does not claim

Contribution, stated plainly. This is a formalization + organization note, not new mathematics. Each ingredient is classical: the autocorrelation theorem (Wiener–Khinchin) and its crystallographic reading — autocorrelation = Patterson, $|\hat{A}|^2$ = diffraction intensity, Z-relation = phase problem — are classical (Patterson 1934 [4]; the Z-relation $\leftrightarrow$ Patterson correspondence is Mandereau et al. 2011 [6]). Homometry is the classical “same interval content” equivalence (Lewin [13]; Soderberg [14]). The DFT-of-pc-sets program is Amiot [7]; the sum / index vector and its relation to $\hat{A}^2$ are within that program — we claim **formalization only**, never priority. The combinatorial sum-vector identity is classical (Lewin [13]; Rahn [15]). DFT injectivity is just that `ZMod.dft` is a `LinearEquiv` [16]. The contribution is the **unified taxonomy as one statement** — pinning each classical invariant to its exact slice of $\hat{A}$ and machine-checking the **two boundaries of the ladder side by side**: the phase-blind collapse (`homometric_iff_normSq_dft_eq`) and the complete-invariant capstone (`Ahat_injective`) in one file with a clean axiom audit. This is the first formalization, and the first time the boundaries are checked together.

What this note does *not* claim.

- **Not new mathematics.** First *formalization* and *organization* only.
- **Not a Lean proof of set-CLASS completeness.** `Ahat_injective` proves $\hat{A}$ determines

the *set*; completeness modulo the T/I phase symmetries is the *computational* statement (Sage sweep, 0 collisions over 224 T/I orbits), not a Lean theorem.

- **Not “full phase” for the sum vector.** $\hat{A}^2$ retains *partial* phase; the Z-pair’s two sum vectors are permutations of each other, so the distinction is indexed (per axis), not a global multiset difference.
- **Not priority over the DFT-of-pc-sets program.** Patterson, Lewin/Soderberg homometry, Amiot’s Fourier program, and Tymoczko–Yust [8] each predate this; we cite, formalize, and do not supplant. The Tymoczko–Yust phase $\approx$ pc-sum result is *under transposition*, a *different* theorem, cited for intuition only.
- **Not a Fourier depth claim about 6-30 self-duality.** $T_6 \in \text{Stab} \iff \hat{A}$ vanishes on odd frequencies is machine-witnessed (0 mismatches over 4096 subsets), but self-duality also needs an inversion-free phase condition $|\hat{A}|$ cannot see — refuted as a clean $|\hat{A}|^2$ theorem in Note #1. **Open frontier:** a Lean theorem for set-class completeness, and lifting the even-frequency-support characterization to general even universes.

Beyond $\mathbb{Z}_{12}$. When the triple correlation ceases to be a complete invariant for general $\mathbb{Z}_n$ — a question of finite additive combinatorics rather than music theory — is taken up in a companion note.

Data and code availability

The Lean source `Fourier.lean` and the Sage witness scripts of Tables 2–3 accompany this note; each Lean theorem is checkable by `#print axioms` and each script is re-runnable.

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